

Toward a Gamification Model to Improve IT Service Management Quality on Service Desk

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Abstract—Among the different kinds of incident processing centers, the service desk has highlighted with the increasing number of its adopters around the world. Nonetheless, even though it has a series of desirable features and resources, a service desk needs several procedures to survive, without which it can severely suffer with inadequate or inexistent IT service management. In order to overcome this need, the framework of best IT practices, mostly known as ITIL, has been widely implemented on such scenarios. However, its adherence by service desk operators, has not always been as friendly as it should. The main reason for this can be made evident when considering how hard is to comply with so many recommendations laid out on such framework. Additionally, the technical staff attention and worry on complying with that framework is so great that incident treatment quality can be impaired. It is exactly on that challenge, where the persuasive technique named gamification arises here as a proposal to turn hard, tense or tedious tasks, that are commonly performed on this way on service desks, into more interesting and engaging activities in a daily basis. Through game design and elements, gamification can improve the service desk operators' motivation and engaging, by making them more involved in their job and consequently optimizing the use and adherence of ITIL recommended IT best practices. In this scenario, since the more IT best practices are adequately followed and the more incident treatment quality increases, obviously the more IT service management quality would be improved.

Keywords—gamification; ITIL; service desk

I. INTRODUCTION

The notable qualities of service desk, including its amplitude of service providing, has made it highly utilized in a huge amount of worldwide organizations. Generally, Information Technology Infrastructure Library - ITIL, that is a well-known IT best practices framework, has been implemented on that kind of incident processing center with

the purpose to optimize and organize its process and activities, as well as support the IT service management.

Even though that framework has notable advantages, it brings some challenges too. It was found out that ITIL implementation can cause an important organizational cultural change, consequently provoking oftentimes, technical staff resistance to adhere and follow the recommended IT best practices proposed by this aforementioned framework [1].

By the way, Brazilian public organizations generally choose just the appropriate IT best practices that can be adequate for its needs. However, most commonly they commit an important mistake, when not adequately adapting the IT best practices proposed by ITIL framework on its real scenario of use. In other words, even Brazilian public organizations choose the appropriate IT best practices to be implemented, they usually fail to adequately evaluate its context and adequately adapt such IT best practices to its real scenario.

Beyond that, as long as the technical staff is ascribed the responsibility of complying with the IT best practices procedures and rules, these can fully be complied, but in many cases, in detriment of incident treatment quality, which are not covered in all aspects measured by the key performance indicators, embedded in the ITIL framework. This scenario can result in a compromised IT service management, in terms of quality.

In this context, it came to surface the whopping need to research some methods or techniques that can minimize those troubles or even solve them. That was the reason which motivated this study to investigate the feasibility of using the persuasive technique so called gamification to attenuate the cultural change impact, as well as its reflex in the incident treatment quality.

Considering the aforementioned necessity of either having a method or technique, to attenuate the cultural change impact, as well as its reflex in the incident treatment

quality, this paper focuses on evaluating the feasibility on applying gamification along with ITIL, on service desks.

That persuasive technique has been implemented in a huge amount of different areas, where in many of them it has reached significant successful outcomes. In that sense, the gamification application spotlight has been directed to areas like education, personal life, government, telecommunications, airlines, location-based services, software testing, and other business, including the so wide are of marketing [2].

Nevertheless, it was found out just a few researches on gamification applied on service desks. Also, even in the broadest area of incident processing centers, researches on gamification are highly scarcity. Still on this bias, however, gamification technique has been already successfully implemented on several organizational areas, in order to improve staff motivation and engaging, but when talking about such persuasive technique application on the specific area of service desk, it is still an innovative area of research.

In this sense, it is important to develop future researches that can more deeply investigate and propose models to overcome the need of more motivation and engaging on service desks using ITIL, as this article has found out through opinion survey applied on a service desk.

Considering such scenario, the strategy adopted here is to lean on both literature review and empirical opinion captured through an opinion survey applied to service desk operators. Then, based on such strategy, this is an initial work going toward a gamification model to improve IT service management quality on service desk.

As a consequence, the main goal of this paper is to investigate the feasibility and description on how the gamification technique can be applied along with ITIL on service desk to attenuate the inherent impacts caused by such application and consequent organizational cultural change.

II. WHAT IS GAMIFICATION?

Through both empirical and theoretical scientific researches, it was pointed out that games have the potential to attract people attention and their efforts, engaging them in a ludic context [3, 4, 5].

Along with society evolution, several new demands have come together, requiring the people being more and more engaged on their activities, doing them with more quality and efficiency. In order to meet this need, it was identified that some game aspects could be extracted to other areas, other than purely games, where engagement is a foible [6, 7].

Consequently, gamification has been adopted in several different areas and sciences, like marketing, education, industry, economy, technology, and frequently on organizational environment [8, 7, 3, 5].

In this bias, it was outlined the game aspects that are direct involved in attracting human attention, identifying as game elements and game designs. These are present at any kind of game, either virtual or real. In other words, they constitute the game main structure [8, 9].

By definition, gamification can be understood as the use of game elements and game designs in non-game contexts, involving challenges, competition, rivalry, being all of them

controlled by well-defined limits and rules, having well-outlined goals [3].

A. Game Elements

The game elements are the inherent characteristics and frequently found out in games, but not necessarily in all of them. They are easily associated with games and have significant importance in game play [5, 8].

For the illustration purpose, it was identified the ten elements more commonly found in the great games, namely: self-representation with avatars; three-dimensional environments; narrative context; feedback; reputations, ranks, and levels; marketplaces and economies; competition under rules that are explicit and enforced; teams; parallel communication systems that can be easily configured; and time pressure [10].

B. Game Designs

By the other side, game designs have a level of abstraction more elevated than game elements, so that its definition is not so simple. Broadly, it is defined as game model, involving its methodologies, characteristics and principles [3].

In order to categorize the game design, it was subdivided into five levels. For materialization purpose, each level has a respective example, as follows:

- Game Interface Design Patterns – badges, leaderboard and levels.
- Game Design Patterns and Mechanics – time constraint, limited resources and turns.
- Game Design Principles and heuristics – enduring play, clear goals and variety of game styles.
- Game Model – challenge, fantasy, curiosity and game design atoms.
- Game Design Methods – playtesting, playcentric design and value conscious game design.

III. IMPLEMENTATION PROPOSAL

A. Delimiting the Scope

As the proposal to implement ITIL along with gamification it is drawn a structure in which the persuasive technique will provide sense and purpose to the stakeholders on complying with every rule and procedure instituted by the ITIL framework in the service desk environment.

Eberlein and Said in [11] has recently proposed a model to implement gamification in the wide are of incident processing centers, where the service desk is situated. Since service desk is the one specifically treated here, assume it as the environment where such model can be implemented.

In parallel, since the more generic term incident processors used in the referred model are equivalent to the more specific term service desk operator, that last one will be used from now on.

In this bias, such gamification model can bring a system based on incident relevance levels, pondering their difficult

and importance as well as the operator skill levels involved on each incident to be treated.

Cartlidge in [12] describes the ITIL *framework* in details, however, for this paper purpose, it will be used here just the service support, which dictates several processes and activities as well as roles that are to be incorporated in service desks.

B. Gamification System Operation

In this context, it is drawn a scenario on which as soon as an incident arises, even though the monitoring system or via client request, the gamification system has to be capable to identify the required skills and competences involved in the incident. From then on, it can point out or assign that incident to the corresponding service desk operator, who should have such needed skill and competences to treat the referred incident [11].

In this process, both game elements and designs are brought together on the job daily activities. When utilizing game elements in the gamified scenario, it has the main purpose to increase operator's motivation [13, 9].

Meanwhile, it is important to note that those game elements have to be implemented within the game design proposed to the service desk specific scenario [14].

As long as an incident demands more and more different skills, competences or specific technical expertise, other operators can contribute toward trouble resolution. Each those possible participations and contribution will be recorded by the gamification system.

Additionally, it is also possible to include some criterion to aggregate more value on rewards. For instance, if a service desk operator solves an unprecedented problem, which until that moment was not present on knowledge base his or her reward has to be greater than another that solved a known problem already documented on knowledge base.

As soon as an incident is solved and its related ticket is closed, depending on the incident difficult and the relevance it represents, a lower or higher reward can be allot to the service desk operator. The rewards can assume different aspects, either internally to the gamification system, like points, levels and badges, as well as externally to it, like social recognition, reputation, and job and salary promotion [15].

Through this process, it is kept a ranking made up by each service desk operator and his or her respective points, levels, progress and badges, stating the accomplishments achieved. As the incidents are solved and their tickets are closed, that ranking is updated with service desk operator recent performance, keeping this way, a historic job profile up to date.

At any given time, the service desk operators as well as supervisors and managers are able to view the individual job profile in detail, as well as the individual situation compared with his or her coworkers.

Based on this individual progress, the gamification system would be able to ponder which kind of incident to assign for each of its service desk operator.

As a criterion to take such decision, the gamification would use the technology expertise, considering the level

and its progress, besides the consideration that every incident should always offer a challenge to the operator working on it. In other words, an incident has to be assigned to an operator, just in case it is in the same or upper level than operator's one. This is a way to keep staff motivated through challenges.

C. Attenuating the Impact Caused by Cultural Change on Service Desk

By incorporating gamification technique along with ITIL on service desk as just proposed, it will create a more interesting and engaging environment, where the service desk operators will face the tasks and activities as challenges and opportunity to increase their ability, organizational visibility and recognition.

In this implementation, the gamification rules and goals have to be properly aligned with service support IT best practices recommended by ITIL framework. In order to align such persuasive technique with that framework, it can be done through correlating the gamification rules and goals with each aspect of ITIL best practices integrated on the service desk.

Those correlations have to be planned in a such way that as a service desk operator fulfill the rules and meet the goals related to an incident on such gamified environment, it corresponds to perfectly comply with ITIL derived procedures, which are inherent to each challenge faced by the service desk operator.

In this way, there is a consequent tendency to have service desk operators making more effort to achieve successful on each incident, following strictly incident rules and procedures recommended by ITIL. Through this engaging and motivating environment, the ITIL impact is attenuated, making the organizational cultural change more slightly than it would be without such persuasive technique.

D. Improving Incident Treatment Quality

As soon as the service desk operators begin to work with ITIL on a gamified service desk as proposed here, naturally more attention will be paid on each incident, including the punctuality with the timeline, update quality in describing the events, accuracy and exactness on information given.

The main reason for this resides in the fact that after each incident has been solved, an evaluation has to be done, analyzing several quality aspects involved in the incident treatment, in order to allot rewards proportionally to each service desk operator that worked on the incident.

Since the service desk operators should work on a competitive and engaging service desk and incident treatment quality is a crucial aspect in the process of individual job performance evaluation, surely more effort will be invested on incident treatment quality on a daily basis.

E. Improving IT Service Management Quality

On both attenuating the ITIL impact on service desk and improving the incident treatment quality on service desk, the gamification technique will consequently improve the IT service management quality as a whole.

In this context, the service desk operators should work highly engaged and motivated, treating the incidents with their best effort, complying with the ITIL recommended best practices. By doing so, the IT service management quality will be improved, since it will be possible to assess each individual job performance as well as team job performance.

Alongside, since the incident treatment quality is an important criterion of evaluation, and the service desk operators should pay more attention on that, the IT service management quality will be positively impacted. This improvement can be made evident when analyzing that IT service management will gain qualitative improvements, more explicitly in the incident treatment quality, rather than its usually measurement as just quantitative.

IV. METHOD

A. Sample Profile

The opinion survey was applied on the public organization named *Núcleo de Tecnologia da Informação – NTI*, which provides and supports IT services to the *Universidade Federal de Alagoas – UFAL*, that is a federal university in Brazilian state *Alagoas*.

This opinion survey was undergone on the service desk. The opinion survey subject were just 8 (eight) service desk operators from such organization, that have partially implemented ITIL framework. These operators have a minimum age of 18 years-old, maximum of 35 years-old and an average of 27 years-old. In this sample, eight out of nine were male sex. The minimum experience time was 1 year or less with 44%, whereas the maximum experience time was 4 years or more with 33%.

B. Proceedings and Instruments

The questions were developed based on the authors experience along with literature review. They were divided into three blocks, each one covering the main aspects of the research.

The first is about how were the current motivation, engagement and satisfaction working with ITIL on service desk, whereas the second was about how was the operators predisposition of working on a more motivating and engaging environment. The third main aspect is around the operators' knowledge and perception about the importance of ITIL on service desk and about the incorporation of new motivating and engaging mechanisms on such environment.

This questionnaire was made up of 24 questions, from which 8 questions makes part of the first block, 7 questions makes part of the second block and 9 questions makes part of the third block. Each question was in the multiple choice format, having 4 options in a scale going from "none", "little", "moderate" and "quite".

Those questions were developed in order to acquire the service desk operators' opinion about each of the aforementioned matters. It was made available to the service desk operators through Google Drive form, during just one day and at their working hour.

V. EVALUATION

A. Block 1 – Motivation, Engagement and Satisfaction

In the block 1, the questionnaire could capture more expressive answers on three questions, which are directly connected to the hypothesis raised on this paper, representing adequately essence of investigation proposed by this block of questions. Those questions were numbered as 4, 12 and 13.

In the question 4, which asked how was the motivation in the professional activities on the service desk, nobody answered "None", just one service desk operators answered "Little", corresponding to 11%, whereas four operators answered "Moderate", corresponding to 44%, tying in number with the other four operators who answered "Quite", as follow the graph below.

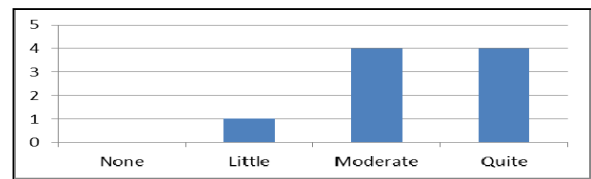


Figure 1. Question 4

As it can be seen in the figure above, 44% of inquired service desk operators answered "Moderate", this can be considered an expressive number in such sample. It is empirical evidence that motivation can still be improved on that scenario.

The question 12 asked the satisfaction on performing the professional activities. Nobody answered "None", but three service desk operators answered "Little", corresponding to 33%, tying in number with the other ones that answered "Moderate" and "Quite", as follows the graph below.

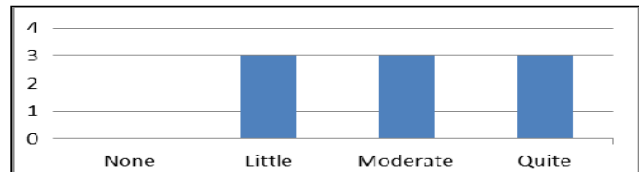


Figure 2. Question 12

As those question answers show, 66% of inquired service desk operators consider themselves with little or moderate satisfaction with their professional activities. This is notable evidence that job satisfaction can still be improved on that scenario.

Question 13 asked to evaluate the engagement on the professional activities. Nobody answered "None", two service desk operators answered "Little", corresponding to 22%, whereas other two answered "Moderate", corresponding to 22%, and five service desk operators answered "Quite", corresponding to 56%, as graph below.

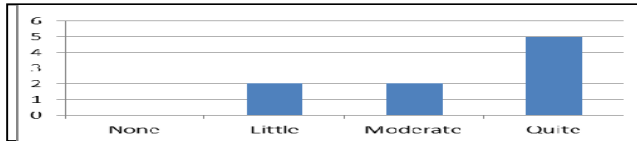


Figure 3. Question 13

That question showed that 44% of inquired service desk operators consider themselves with little or moderate engagement to deal with professional activities. This evidences that engagement still can be improved on that scenario.

B. Block 2 – Predisposition to Face Changes

In the block 2, the questionnaire could capture more expressive answers on two questions, which are directly connected to the hypothesis raised on this paper, representing adequately essence of investigation proposed by this block of questions. Those questions were numbered as 6 and 7.

Question 6 asked to evaluate the importance of incorporating adequate challenges in the daily professional activities. In this question, nobody answered “None” neither “Little”. Two service desk operators answered “Moderate”, corresponding to 22%, and the other seven service desk operators answered “Quite”, corresponding to 78%, as can be seen in the graph below.

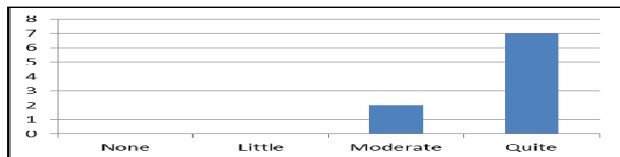


Figure 4. Question 6

This question presents a large majority of 78% of the inquired service desk operators considering quite important the incorporation of adequate challenges in the professional activities. This is notable evidence that there is a positive predisposition to face changes in that scenario.

Question 7 asked to evaluate the benefits of becoming the service desk more ludic. Nobody answered “None” or “Little”, three service desk operators answered “Moderate”, corresponding to 33%, and the other six service desk operators answered “Quite”, corresponding to 67%, as showed in graph below.

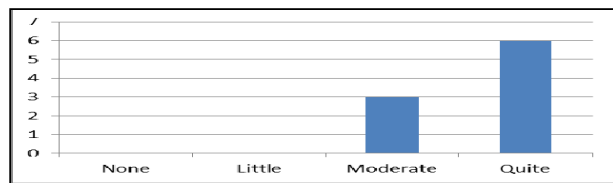


Figure 5. Question 7

In this question, it was possible to capture in the opinion informed the tendency to face change with a positive predisposition, the majority of inquired service desk

operators did opine that becoming the service desk more ludic is quite benefic.

C. Block 3 – Knowledge and Comprehension of Importance

In the block 3, the questionnaire could capture more expressive answers on two questions, which are directly connected to the hypothesis raised on this paper, representing adequately essence of investigation proposed by this block of questions. Those questions were numbered as 2 and 21.

Question 2 asked to evaluate the importance of following ITIL framework best practice in the incident treatment. Nobody answered “None”, just one service desk operator answered “Little”, corresponding to 11%, whereas one service desk operator answered “Moderate”, corresponding to 11%, and the other seven service desk operator answered “Quite”, corresponding to 78%, as graph below.

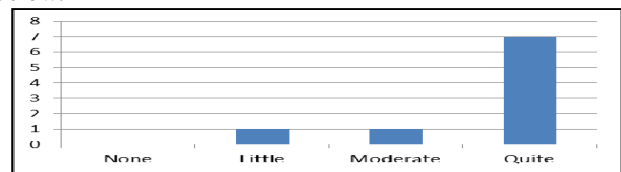


Figure 6. Question 2

In this question, it was possible to identify that the majority of inquired service desk operators understand that following the ITIL framework best practice in the incident treatment has a quite importance. This can evidence that ITIL is considerably comprehended by the service desk operators in this scenario.

Question 21 asked to evaluate the necessity to stablish mechanisms that can promote more attention of coworkers and managers, making possible more visibility and consequently more recognition based on performance. Nobody answered “None” or “Little”, but three service desk operators answered “Moderate”, corresponding to 33%, whereas six service desk operators answered “Quite”, corresponding to 67%, as can be seen in the graph below.

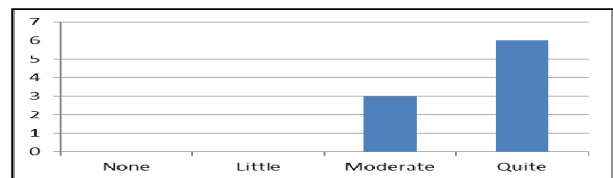


Figure 7. Question 21

This question could capture in the opinion informed that inquired service desk operators majority has a comprehension about the need of implementing mechanisms that can make the job performance more transparent. This is an evidence that service desk operators comprehend the benefits and importance of making the job performance more transparent.

VI. FINAL CONSIDERATIONS

This paper dedicated on studying the feasibility of implementing persuasive technique along with ITIL on service desk. This study leaned on literature review as well as empirical research, through an opinion survey applied on a Brazilian public organization's service desk.

Based on the literature review, it was proposed an initial step toward a gamification model to improve IT service management quality on service desk. In this initial step, it was argued how the persuasive technique gamification can contribute to increase motivation and engagement on a service desk with ITIL implemented, and consequently improving the IT service management quality.

By evaluating the opinion survey, it was possible to identify important empirical information related to the feasibility on implementing the persuasive technique along with ITIL on service desk.

This empirical opinion survey served just as an example to reinforce the hypothesis raised here about the necessity of implementing gamification on such environment to improve IT service management quality.

At this point, an important research goal was achieved, since such opinion survey revealed that on the service desk investigated, there is considerable need to improve motivation, engagement, recognition, satisfaction and competitiveness on the referred scenario. Beyond that, it was also possible to identify through the opinions informed a crucial positive predisposition to face changes on such scenario.

As the paper result, it presents some evidences about a possible feasibility for gamification models to be implemented along with ITIL to improve IT service management quality on service desk. Those evidences point out the necessity to improve motivation and engaging on service desk operators that uses IT best practices recommended by ITIL framework, on their daily activities. Certainly, those evidences can also indicate that further and deeper researches have to be carried out to either study the mentioned aspects and others, either even to propose models that can attend such demand.

Along with the aforementioned result, some important limitations were brought up on this research. As an important limitation, it can be taken into consideration the little amount of service desk operators in the sample that answered the opinion survey. This limitation was caused by lack of financial resources along with lack of further Brazilian enterprises collaboration. Beyond that, another relevant limitation that was raised during the research was time constraint, which prevented this article to include another and broader questionnaire, approaching and evaluating other involved aspects, besides of applying it on a larger sample.

As a proposal to implement gamification along with ITIL on service desk, it was drawn a structure in which the persuasive technique provides sense and purpose to the service desk operators on complying with ITIL framework, benefiting the IT service management quality.

This can happen through the increasing of motivation, engagement, satisfaction, competitiveness and transparence

for recognition, which are all of them, in the gamification focus and potential.

In the investigation process, different aspects of such implementation was approached, expanding the discussion for further investigation and future researches in related the area.

Therefore, for the next research, it is intended to produce an article that studies a gamification model implementation on a service desk along with ITIL and evaluate before and after that implementation how motivation, engaging, and IT service management quality can be affected. In order to evaluate those criterions, a broader questionnaire covering motivation, engaging, and IT service management quality has to be applied on a large sample of service desk operators, both before and after the gamification model implementation.

REFERENCES

- [1] A. C. Steel, "Transforming IT service management – the ITIL impact," 17th Australian Conference on Information Systems, 6-8 Dec 2006, Adelaide, Australia.
- [2] B. M. Paz, "Gamification: a tool to improve sustainability efforts," A dissertation submitted to The University of Manchester for the degree of Master of Science in the Faculty of Engineering and Physical Sciences, 2013, Manchester, England.
- [3] S. Deterding, D. Dixon, R. Khaled, L. Nacke, "From game design elements to gamefulness: defining "gamification"," Mindtrek'11, September 28-30, 2011, Tampere, Finland.
- [4] Chourney, A. (2012) Taking The Game Out Of Gamification. *Dalhousie Journal of Interdisciplinary Management*.
- [5] Groh, F. (2012) Gamification: State of the Art Definition and Utilization. *4th Seminar on Research Trends in Media Informatics*.
- [6] Fitz-Walter, Z.; Tjondronegoro, D. W. & Wyeth, P. (2011) Orientation passport : using gamification to engage university students. In *Proceedings of the 23rd Australian Computer-Human Interaction Conference, ACM*, Australian National University, Canberra, ACT.
- [7] Huotari, K., & Hamari, J. (2011). "Gamification" from the perspective of service marketing. *CHI'2011 (Gamification workshop)*, Vancouver, Canada.
- [8] Landers, R. N. & Callan, R. C. (2011). Casual social games as serious games: The psychology of gamification in undergraduate education and employee training. In M. Ma, A. Oikonomou, & L. C. Jain (Eds.), *Serious Games and Edutainment Applications* (pp. 399-424). Surrey, UK: Springer.
- [9] Steinkuehler, C. (2008). Introduction to the Games Learning & Society (GLS) Conference Special Issue. *Games and Culture*, vol. 3, n° 3-4, pp. 251-252.
- [10] I. Blohm, J. M. Leimeister, "Gamification. Design of IT-based enhancing services for motivational support and behavioral change," *Business & Information Systems Engineering (BISE)*, 2013, Erscheinungsjahr, Germany.
- [11] P. Eberlein, B. Said, "Applying gamification techniques to process incidents," Patent Application Publication, Dec 5, 2013, United States.
- [12] A. Cartledge, "An Introductory Overview of ITIL ® V3," The UK Chapter of the itSMF. 2007
- [13] Shneiderman, B. (2004) Designing the user interface: Strategies for effective Human-Computer Interaction. 4. ed. Maryland, Addison-Wesley Publishing Company.
- [14] Von Ahn, L., & Dabbish, L. (2008). Designing games with a purpose. *Communications of the ACM*, 51(8), 58-67.
- [15] S. Lee, *The multiplayer classroom: Designing coursework as a game*. Cengage Learning, 2012.